Kaplan-Meier Subgroup Survival Analysis

Notebook: `03-survival_KM_subgroups.ipynb`
Date: April 21, 2026
Outcome: Overall survival (OS), truncated at $\tau = 57$ months
Dataset: `GPT_processed_survival_data.csv`

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1. Study Sample

Parameter	Value
Total N	264
Death events	101 (38.3%)
Truncation time τ	57 months (inherited from `03-survival-kaplan_meier.ipynb`)

Subgroup analyses were performed for **Sex**, **Tumor type**, and **Age** (continuous and tertile groups), all using truncated survival times and events.

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2. Sex

2.1 Group Summary

Sex	n	Events	Event rate
0 (female)	111	47	42.3%
1 (male)	153	54	35.3%

Both groups meet the >= 10 events reliability threshold.

2.2 Log-rank Test

Test statistic	p-value
1.48	0.2234

No statistically significant difference in survival between sexes ($p = 0.22$).

2.3 Cox PH Model (univariable, Sex coded)

Proportional hazards assumption: **satisfied**.

The Cox model confirms no significant sex effect on survival. The HR estimate direction is consistent with the slightly higher event rate in females (Sex = 0), but the difference is not significant.

2.4 Power Assessment (Schoenfeld)

Comparison	d	HR	Estimated power
Female vs. Male	101	0.83	14.5%

WARNING Power is well below the 80% threshold. The study is **underpowered to detect the observed sex effect** (HR ~ 0.83). The non-significant log-rank result should be interpreted cautiously — a true moderate sex effect cannot be ruled out.

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3. Completeness of Cytoreduction (CC)

3.1 Group Summary

CC	n	Events	Event rate	Events >= 10?
0 (complete)	179	65	36.3%	OK
1 (incomplete)	84	35	41.7%	OK

Both groups meet the >= 10 events reliability threshold. Total N = 263; total events = 100 (38.0%).

3.2 Log-rank Test

Test statistic	p-value
2.90	0.0888

No statistically significant difference in survival between CC groups at the 0.05 level (p = 0.089), though the trend is in the expected direction (higher event rate with incomplete cytoreduction).

3.3 Power Assessment (Schoenfeld)

Comparison	d	HR	Estimated power
CC0 vs CC1	100	1.15	9.4%

WARNING Power is well below the 80% threshold (9.4%). The study is **severely underpowered to detect the observed effect size** (HR ~ 1.15). The non-significant log-rank result is consistent with low power rather than a true null effect. Note that in multivariable analyses (see `04-cox_survival_report.md`), CC is a significant predictor of OS (HR = 1.67, p = 0.022) after adjusting for tumour type and age — the univariable signal is suppressed here by the small absolute HR and limited events per group.

4. Tumor Type

4.1 Group Summary

Tumor	n	Events	Event rate	Events >= 10?
1	99	48	48.5%	OK
2	18	7	38.9%	WARNING Low
3	22	6	27.3%	WARNING Low
4	48	8	16.7%	WARNING Low
5	23	5	21.7%	WARNING Low
6	31	18	58.1%	OK
7	13	4	30.8%	WARNING Low
8	10	5	50.0%	WARNING Low

Only tumor types 1 and 6 have >= 10 events; all other groups have sparse event counts.

4.2 Multivariate Log-rank Test

Test statistic	p-value
48.08	3.43×10^{-8}

Tumor type is a **highly significant predictor of survival** ($p < 0.001$). The KM curves show clearly differentiated survival trajectories across tumor groups.

4.3 Pairwise Power Assessment (Schoenfeld)

Only **two pairwise comparisons** meet the >= 80% power threshold:

Comparison	d	HR	Estimated power	Conclusion
T1 vs T4	56	0.34	97.9%	OK Sufficient power
T1 vs T5	53	0.45	83.1%	OK Sufficient power
T4 vs T6	26	3.48	88.9%	OK Sufficient power

All other pairwise comparisons are underpowered (< 80%), primarily due to low event counts in most tumor-type subgroups. The overall multivariate test is robust (large combined events), but **individual pairwise contrasts should be interpreted with caution** except for the three adequately powered comparisons above.

The patterns are consistent with the Cox PH results from notebook `04-cox_standard.ipynb`: tumor types 3, 4, 5, and 7 show markedly lower event rates than type 1, while type 6 shows the highest event rate of all groups (58.1%).

5. Age

5.1 Tertile Groups (KM curves)

Age was divided into tertiles (Younger / Middle / Older) for visual inspection of KM curves. No formal test was performed on the tertile groups; the Cox PH model on continuous Age is the primary inferential analysis.

5.2 Cox PH Model (univariable, Age continuous)

Parameter	Value
Total N	264
Total events	101 (38.3%)
Age mean	54.2 years
Age SD	11.6 years

HR per 1-year increase	95% CI	Wald p-value
0.991	[0.973, 1.008]	0.287

Age is **not a significant predictor** of overall survival (p = 0.287). The HR per year is essentially 1.0.

Proportional hazards assumption: **satisfied**.

5.3 Power Assessment (Schoenfeld, per SD of Age)

log(HR) per SD	Estimated power
0.111	8.1%

WARNING Power is extremely low (8.1%). The study has **insufficient power to detect the observed age effect** (HR ~ 0.99 per year, equivalent to HR ~ 0.90 per SD). A clinically meaningful age effect cannot be excluded on statistical grounds alone.

5.4 Predicted Survival Curves

Cox-predicted survival curves were plotted for the P25, P50, and P75 percentiles of Age, confirming the near-flat effect: the survival curves at different age values are very close across the follow-up period.

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6. Summary

Variable	Test	Test statistic	p-value	Significant	Power adequate?
Sex	Log-rank	1.48	0.2234	No	No (14.5%)
CC	Log-rank	2.90	0.0888	No (trend)	No (9.4%)
Tumor type	Multivariate log-rank	48.08	3.43×10^{-8}	**Yes**	Overall yes; pairwise mostly no
Age	Cox Wald test	—	0.287	No	No (8.1%)

Key conclusion: Tumor type is the only subgroup variable with a statistically significant association with overall survival in this univariable analysis. Sex, CC, and Age are all non-significant at the 0.05 level. However, all three are severely underpowered (power < 15%) for their observed effect sizes, so non-significance should not be interpreted as evidence of no effect. Notably, CC becomes a significant predictor in the multivariable Cox model (HR = 1.67, p = 0.022, see `04-cox_survival_report.md`), where tumour type is controlled for. The univariable CC log-rank result is therefore likely attenuated by confounding with tumour type distribution.